

STATISTICS

Hypothesis Testing Part 2

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ERRORS

Karar	Exact Situation	
	H_0 is True	H_0 is False
Accept H_0	Correct Decision (1 - α)	II.Type Error (β)
Reject H_0	I.Type Error (α)	Correct Decision (1 - β)

1 - α : It states the confidence level of test. It is the probability of accepting H_0 when H_0 is true.

1 - β : The power of the test. It is the probability of rejecting H_0 when H_0 is false.

The Determination of β Error for Population Mean

Consider the two-tailed hypothesis testing:

$$H_0: \mu = \mu_0$$

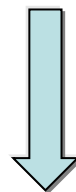
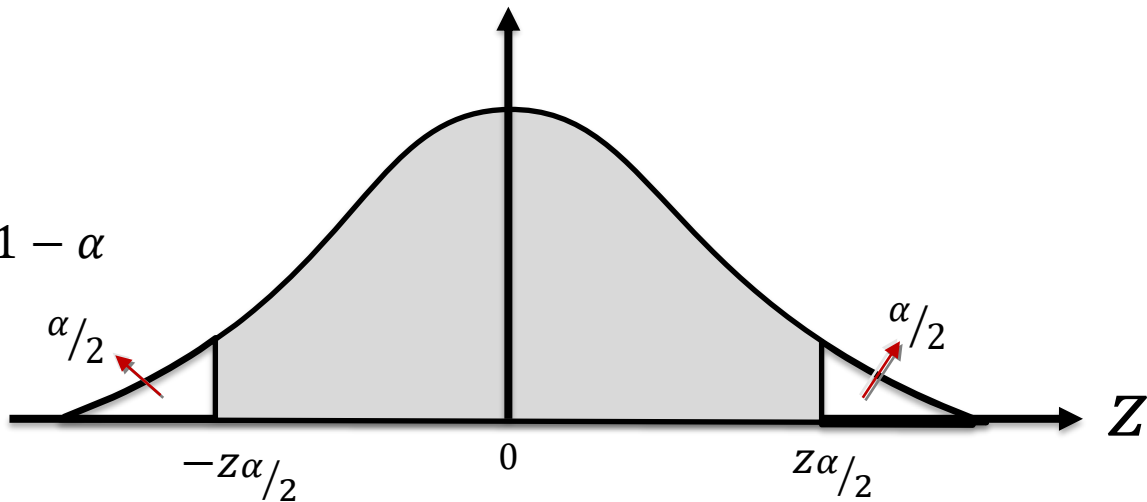
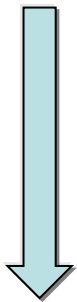
$$H_a: \mu \neq \mu_0$$

Assume that the population has normal distribution with known standard deviation σ . If H_0 is true, then

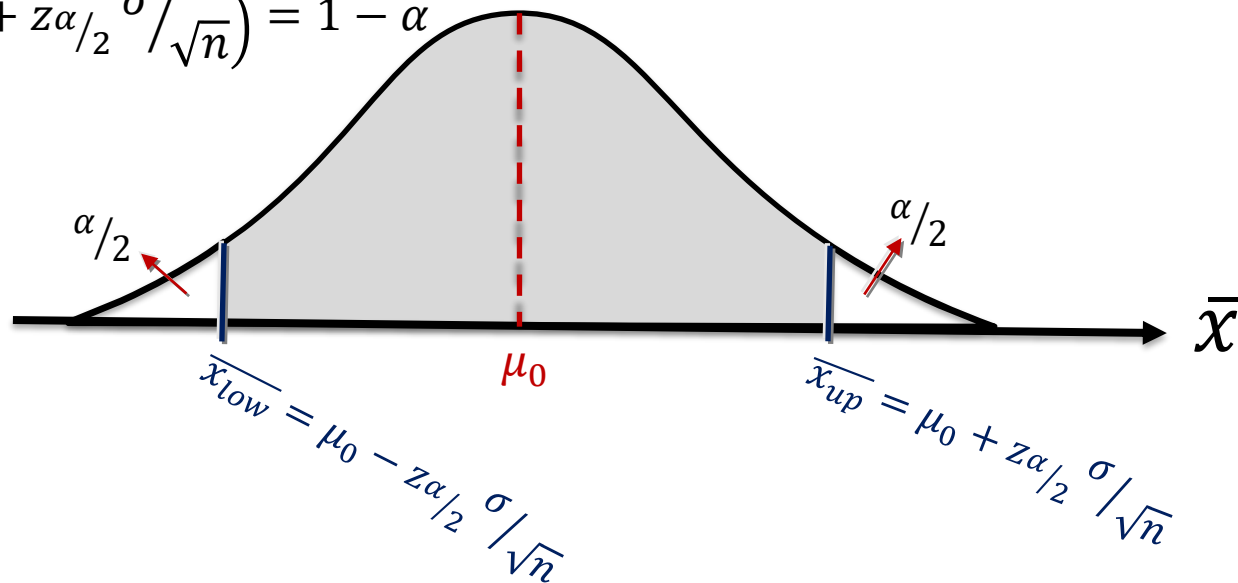
$$P\left(-Z_{\alpha/2} < \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < Z_{\alpha/2}\right) = 1 - \alpha$$

If H_0 is true, then

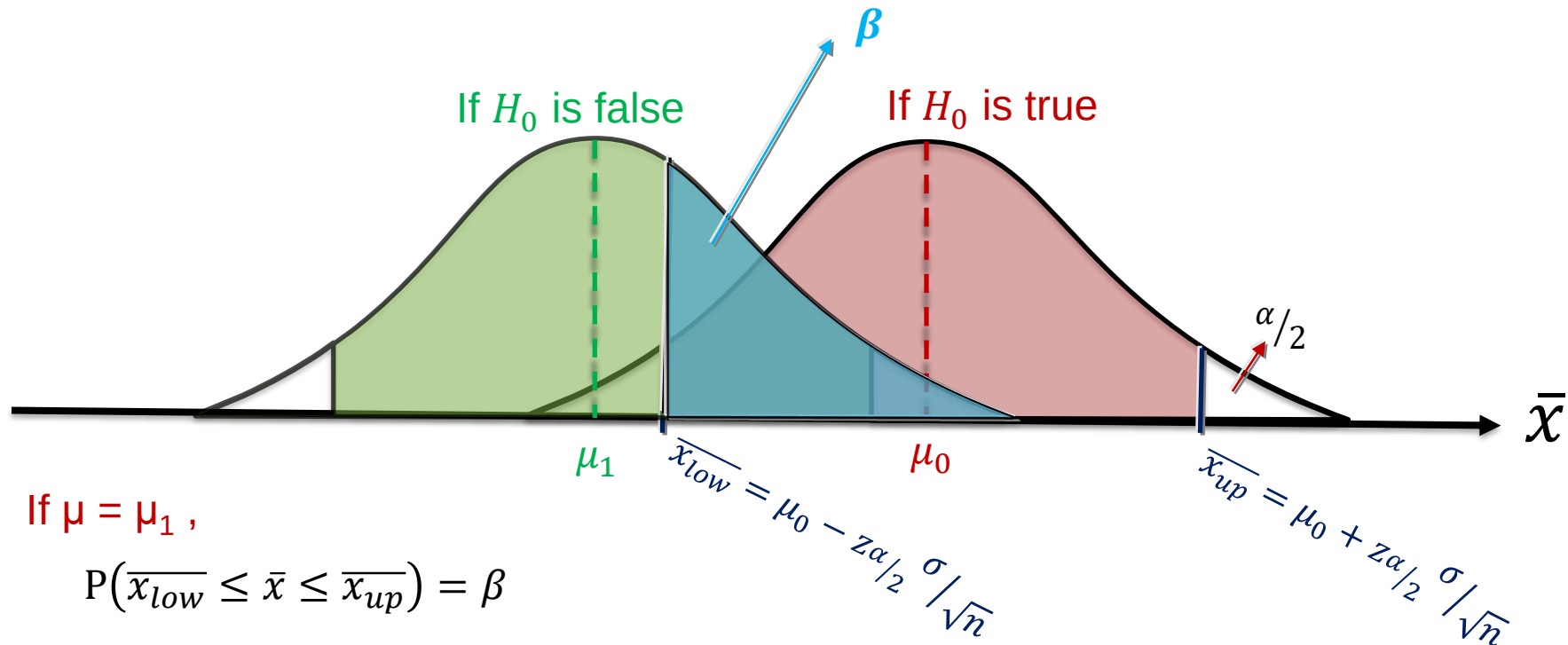
$$P\left(-z_{\alpha/2} < \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$



$$P\left(\mu_0 - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{x} < \mu_0 + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$



If H_0 is false and the exact value of the population mean μ_1 , then ...



If $\mu = \mu_1$,

$$P(\bar{x}_{low} \leq \bar{x} \leq \bar{x}_{up}) = \beta$$

$$\Rightarrow P\left(\mu_0 - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \bar{x} \leq \mu_0 + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = \beta$$

$$\Rightarrow P\left(\frac{\mu_0 - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} - \mu_1}{\sigma/\sqrt{n}} \leq \frac{\bar{x} - \mu_1}{\sigma/\sqrt{n}} \leq \frac{\mu_0 + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} - \mu_1}{\sigma/\sqrt{n}}\right) = \beta$$

$$\Rightarrow P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) = \beta$$

If β is obtained similarly for one-tailed hypothesis, one can write the following probabilities for β :

1) If $H_0: \mu = \mu_0$
 $H_a: \mu \neq \mu_0$ 
$$P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) = \beta$$

2) If $H_0: \mu \geq \mu_0$
 $H_a: \mu < \mu_0$ 
$$P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha} \leq z\right) = \beta$$

3) If $H_0: \mu \leq \mu_0$
 $H_a: \mu > \mu_0$ 
$$P\left(z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha}\right) = \beta$$

Example 1:

A company produces a specific kind of soaps, and it claims that the soap's PH value is less than 5.5. 36 soaps are selected, and it is examined that the average and standard deviation of their PH value is 5 and 1.5, respectively. Test the hypothesis for $\alpha = 0.05$ and then find the power of test for $\mu_1 = 5$

$$\begin{array}{l} H_0: \mu = 5.5 \\ H_a: \mu < 5.5 \end{array} \quad n = 36 \quad \bar{x} = 5 \quad s = 1.5 \quad \mu_1 = 5 \quad 1 - \beta = ?$$

$$z_0 = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{5 - 5.5}{1.5 / \sqrt{36}} = -2 \quad z_\alpha = z_{0.05} = 1.645$$

Since $-2 < -1.645$, H_0 is rejected.

If we accept H_0 when it is false, then

$$P\left(\frac{\mu_0 - \mu_1}{s/\sqrt{n}} - z_\alpha \leq z\right) = \beta$$

$$P\left(\frac{5.5 - 5}{1.5/\sqrt{36}} - 1.645 \leq z\right) = \beta$$

$$P(0.36 \leq z) = \beta \quad \Rightarrow \quad \beta = 0.5 - 0.1406 = 0.3594$$

$$\Rightarrow \quad 1 - \beta = 0.6406$$

Example 2:

Solve the previous example by assuming that the population has a normal distribution with a known standard deviation of 1.5 for $n=9$ and $\mu_1 = 6$.

$$\begin{array}{l} H_0: \mu = 5.5 \\ H_a: \mu < 5.5 \end{array} \quad n = 9 \quad \bar{x} = 5 \quad \sigma = 1.5 \quad \mu_1 = 6 \quad 1-\beta = ?$$

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{5 - 5.5}{1.5 / \sqrt{9}} = -1 \quad z_\alpha = z_{0.05} = 1.645$$

Since $-1 > -1.645$, H_0 is accepted.

If we accept H_0 when it is false, then

$$P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_\alpha \leq z\right) = \beta$$

$$P\left(\frac{5.5 - 6}{1.5/\sqrt{9}} - 1.645 \leq z\right) = \beta$$

$$P(-2.64 \leq z) = \beta \quad \Rightarrow \quad \beta = 0.5 + 0.4959 = 0.9959$$

$$\Rightarrow \quad 1 - \beta = 0.0041$$

DETERMINATION OF SAMPLE SIZE

Consider the determination of β ,

For a two-tailed test,

$$\begin{array}{l} \text{If } H_0: \mu = \mu_0 \\ H_a: \mu \neq \mu_0 \end{array} \quad \Rightarrow \quad P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) = \beta$$
$$\Downarrow$$
$$\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_{\alpha/2} = z_\beta \quad \text{or} \quad \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{\alpha/2} = -z_\beta \quad \Rightarrow \quad n = \frac{(z_\beta + z_{\alpha/2})^2 \sigma^2}{(\mu_0 - \mu_1)^2}$$

For one-tailed tests,

$$\begin{array}{l} \text{If } H_0: \mu \geq \mu_0 \\ H_a: \mu < \mu_0 \end{array} \quad \Rightarrow \quad P\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_\alpha \leq z\right) = \beta \quad \Rightarrow \quad \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} - z_\alpha = z_\beta$$
$$\begin{array}{l} \text{If } H_0: \mu \leq \mu_0 \\ H_a: \mu > \mu_0 \end{array} \quad \Rightarrow \quad P\left(z \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_\alpha\right) = \beta \quad \Rightarrow \quad \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_\alpha = -z_\beta$$
$$\Downarrow$$
$$n = \frac{(z_\beta + z_\alpha)^2 \sigma^2}{(\mu_0 - \mu_1)^2}$$

Note that, during the computation of n , the positive z_β should be considered.

Example 3:

Let $\sigma=315$, $\mu_0=2100$, $\mu_1=2550$, $\alpha=0.05$ and $1-\beta=0.85$. Find the sample size to satisfy the given conditions when applying a two-tailed test.

$$z_{\alpha/2} = z_{0.025} = 1.96 \qquad z_{\beta} = z_{0.15} = 1.04$$

$$n = \frac{(z_{\beta} + z_{\alpha/2})^2 \sigma^2}{(\mu_0 - \mu_1)^2} = \frac{(1.04 + 1.96)^2 315^2}{(2100 - 2550)^2} = 4.410 \sim 5$$